

Extracting morphological characteristics of diseased fingerprints

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1 Introduction

Fingerprints are one of the most common biometrics trait used in practice. This is thanks to their large intersubject variability and wide acceptance among people. However some people suffering from dermatological issues are impaired in use of these biometric systems, because their fingerprints can become too disrupted for recognition. This complication are trying to battle some algorithms that focus on restructuring and enhancing of diseased parts, that allow some damage to be accepted.

This project is focused on morphological extraction of diseased parts from such fingerprints. This process highlights areas to researchers and allows other algorithms to target specified area. For this purpose a few algorithms are studied and tested and based on them new algorithm is implemented.

2 State of the Art

A variety of fingerprint scanner types is in use nowadays. Mainly thermal, capacitance, ultrasonic and optical. Thermal scanners use the heat transferred from contact to differentiate between ridges and valleys of fingerprint. Capacitance scanners work similarly but with electric charge. Ultrasonic scanners use sound waves to penetrate top layer of skin and based on reflection detect papillary lines on the junction of epidermis and dermis. Optical scanners take detailed picture of the finger and detect ridges and valleys based on difference in color.

Diseases that cause histopathological changes in both dermis and epidermis affect all the mentioned types of scanners. Additionally diseased causing discoloration of skin can affect optical scanners and diseases affecting aforementioned junction of dermis and epidermis are a problem for ultrasonic scanners.[2]

2.1 Skin

Skin is an organ that covers body and provides several functions such as sensation, heat regulation, excretion and protection. Skin consist of three layers: epidermis, dermis and fat layer. Epidermis is the outermost layer of skin and is responsible for protection as it prevents pathogens from entering body. Epidermis contains no blood vessels and cells are only supplied by diffusion from capillaries in dermis. Dermis is a layer bellow epidermis which provides skin with elasticity. Dermis contains many nerve endings, sweat glands, hair follicles and is supplied with blood through vessels. Dermis contains wrinkled papillary region which is responsible for existence of ridges and valleys and functions as strong connection with epidermis.

2.2 Fingerprint

Fingerprints are an impression of ridges and valleys mentioned in 2.1 located on the fingertips. They are almost unique among people with estimated probability of 1.9×10^{15} of two fingerprints being the same [7]. Based on basic shape fingerprints are usually categorized to 3 basic categories: Arch, whorl and loop. These categories can be further subdivided for example left and right loop, simple and tented arch or simple, pocketed and accidental whorl [5].

Most common features used for identification are ridge minutiae which consist of two main components: bifurcation and ridge end. More complex minutiae consisting from these two basic patterns can be detected in fingerprints such as double bifurcation, spur, crossing, dot, island, whorl, bridge, etc. [3]. Fingerprint is characterized by presence of these minutiae, their location and their orientation. Fingerprint image usually can contain between 25 to 80 such minutiae [8].

2.3 Disease Detection Algorithms

Several algorithms using fingerprint directional field and histogram analysis methods were proposed before [1, 6]. Directional field method extracts rough angle of ridge lines and other lines from fingerprint in small chunks, every angle is compared against its neighbors and is considered anomalous if it too much differs from the rest. This method should be able to detect absence of ridge lines such as with blisters and lines caused by cuts and scarring. Histogram analysis is a simple method based on observation that most ridge lines and valleys have specific contour in image. Overall histogram of contours is computed and compared against small chunks of image. This method should be capable of detecting absence of ridge lines and discoloration of skin.

This project is aimed at exploration of possibility of extracting diseased parts using morphological functions before used for extraction of fingerprint as a whole [4, 7, 9]. Basic morphological functions are *dilation* and *erosion*. Morphological functions work with kernel called element. Dilation is used to enlarge objects as such every pixel is overlaid with the element and is set to maximal value from neighborhood defined by element. Erosion works similarly but with minimum of neighborhood thus shrinking objects. Due to the nature of these operations they work in single color dimension that means usually gray-scale or binary image. Their effect can be seen in figure 1(b,c).

Advanced morphological operations are made from combining dilation and erosion and include *opening* and *closing*. Opening is erosion followed by dilation, this smooths edges and eliminates peaks and lines. Similarly closing is dilation followed by erosion, which fills small holes and connects shapes. Their effects are again pictured in figure 1(d,e).

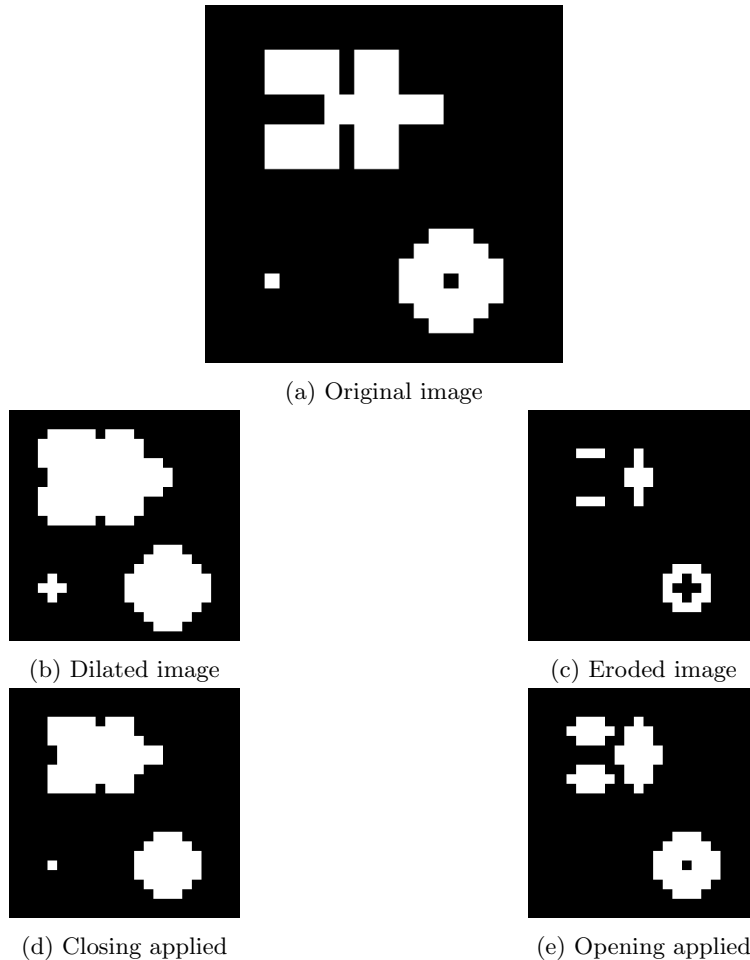


Figure 1: Demonstration of morphological functions with 3×3 cross element

Using morphological functions combined with flood fill and thresholding it should be possible to extract abnormalities such as absence of ridge lines and discoloration, based on the level of thresholding, in form of binary map. Idea is to use closing to merge ridge lines into a single component leaving blank spots from their absence.

3 Experiment

The database of damaged fingerprints was provided by the university. Images were loaded to python using openCV. The image was binarized by using adaptive gaussian thresholding with varying sizes of sliding window and threshold offset, this was done to normalize discoloring of fingerprints caused by difference in pressure applied while scanning. To remove background original image was also simply binarized but this time using mean as threshold, main fingerprint area was merged to single color using morphological closing and outer area was then floodfilled to remove any noise from background. Using acquired background filter noise, which was created by adaptive threshold, was removed from image. Then effect of dilation and erosion with element of different sizes (smaller than a ridge line and bigger than a ridge line) was investigated.

The map visible in figure 2 was acquired by using Gaussian thresholding with large window, then closing with circular element of diameter 11 to merge ridge lines and then opening with same element to polish spot left behind by wart. As it appears bottom part of wart was consumed by dilation of ridge lines.

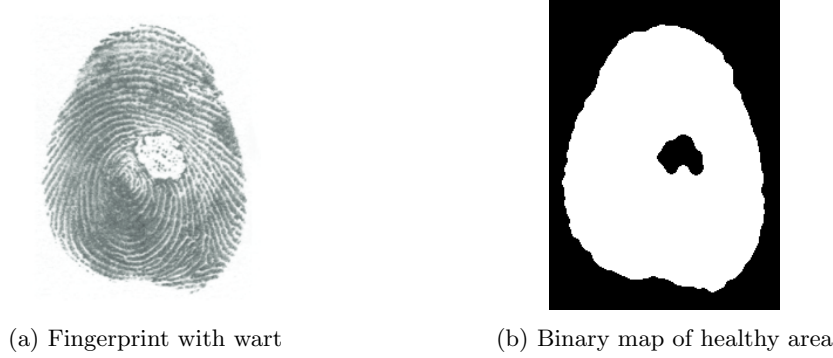


Figure 2: An image of a diseased fingerprint with its binary map constructed by morphological operations

Before flood-fill was used for background recognition the morphological functions were completely used to extract fingerprint area as shown in 3(a). This mask was also eroded with element of diameter 5 to accommodate for later mask joining. This method was found to be lacking due to its dependability on large morphological element, which could cause a significant loss in shape.

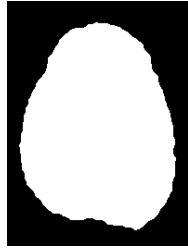
Similar process to one mentioned above was used to acquire wart in image, though with colors inverted. Image is firstly blurred using Gaussian blur of diameter 15 to soften lone dots which caused imperfect detection in 2. Then closing with diameter 11 is applied and image is again thresholded with threshold set to 240. In the end image is dilated by element of size 7 to compensate for radius of applied blur. The second mask can be seen in 3(b). Both masks are combined with bit-wise and operation and area highlighted by them in original image can be seen in image 3(c).

Later use of blur was rejected, because it's function can be completely replaced with morphological functions. By application of dilatation with small element it is possible to remove small noisy black dots. Then by using erosion with bigger element noisy white dots, which grew with dilatation, can be removed. Depending on the wanted level of sensitivity scale of erosion can be increased. Then large dilatation is applied to connect shapes. Lastly erosion is used to scale areas back to their size. Image was split to two parts one containing ridges and the other valleys of fingerprint. This was done to highlight not only pits in fingerprint but also plateaus and to separately control sensitivity and thus false positive rate.

As seen in images 4 usage of morphological functions has several issues. With cuts too narrow this method is unable to recognize cut from valley thus generating false negative. Some normal patterns in fingerprints may also generate false positive, especially forks in ridges seem to be prone to this in plateau detection.

4 Summary

Usage of morphological functions in fingerprint disease recognition brings an easy way to detect and precisely locate large deformations to fingerprint. While their use alone does not warranty successful detection of smaller cuts it should be possible to combine this method with other methods that follow direction of fingerprint ridge lines to enhance its performance.



(a) Binary map of full fingerprint



(b) Binary map of absence of ridge lines



(c) Area denoted by combination of mask maps

Figure 3: Combination of two masks was used to highlight area affected by wart



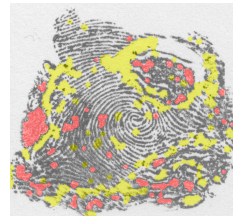
(a) Binarized print generated by thresholding



(b) Fingerprint with algorithm highlighted areas. Plateaus in red, pits in yellow.



(c) Binarized print generated by thresholding



(d) Fingerprint with algorithm highlighted areas. Plateaus in red, pits in yellow.

Figure 4: Binarized image of a diseased fingerprint and areas highlighted by algorithm

References

- [1] BAROTOVÁ, Š. Detector of skin diseases by fingerprint technology. Bachelor's thesis, Brno University of Technology, 7 2017.
- [2] DRAHANSKY, M., DOLEZEL, M., URBANEK, J., BREZINOVA, E., AND HOON KIM, T. Influence of skin diseases on fingerprint recognition. *Journal of Biomedicine and Biotechnology* 2012 (2012), 1–14.
- [3] DRAHANSKY, M., AND KANICH, O. BIO - Rozpoznávání podle otisků prstů, 2022.
- [4] FAHMY, M. F., AND THABET, M. A. A fingerprint segmentation technique based on morphological processing. In *IEEE International Symposium on Signal Processing and Information Technology* (Dec. 2013), IEEE.
- [5] KAUSHAL, N., AND KAUSHAL, P. Human identification and fingerprints: A review. *Journal of Biometrics & Biostatistics* 02, 04 (2011).
- [6] NEMANJA, V. Partial fingerprint detection using blob detection algorithm. Bachelor's thesis, Brno University of Technology, 7 2019.
- [7] SINGH, P., AND KAUR, L. Fingerprint feature extraction using morphological operations. In *2015 International Conference on Advances in Computer Engineering and Applications* (Mar. 2015), IEEE.
- [8] THAKKAR, D. Minutiae based extraction in fingerprint recognition, Jul 2020.
- [9] THIRAN, J.-P., AND MACQ, B. Morphological feature extraction for the classification of digital images of cancerous tissues. *IEEE Transactions on Biomedical Engineering* 43, 10 (1996), 1011–1020.
- [10] VANDAELE, R., NERVO, G. A., AND GEVAERT, O. Topological image modification for object detection and topological image processing of skin lesions. *Scientific Reports* 10, 1 (Dec. 2020).